Example 5.1

Find the unknowns in the following figures

(i)

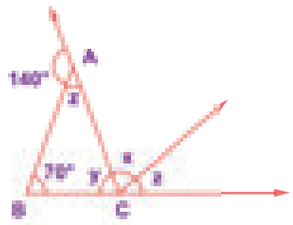


Fig. 5.7(i)

(ii)

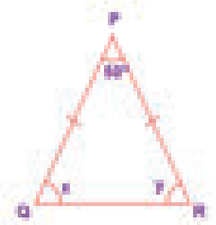


Fig. 5.7(ii)

(iii)

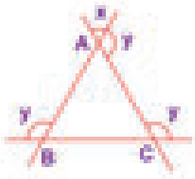


Fig. 5.7(iii)

Solution:

(i) Now, from Fig. 5.7(i), $140^\circ + \angle z = 180^\circ$ (linear pair)

$\Rightarrow \angle z = 180^\circ - 140^\circ = 40^\circ$

Also $\angle x + \angle z = 70^\circ + \angle z$ (exterior angle property)

$\Rightarrow \angle x = 70^\circ$

Also $\angle z + \angle y + 70^\circ = 180^\circ$ (angle sum property in Δ ABC)

$\Rightarrow 40^\circ + \angle y + 70^\circ = 180^\circ$

$\Rightarrow \angle y = 180^\circ - 110^\circ = 70^\circ$

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(ii) Now, from Fig. 5.7(ii), $PQ = PR$

$\Rightarrow \angle Q = \angle R$ (angles opposite to equal sides are equal)

$\Rightarrow \angle x = \angle y$

$\Rightarrow \angle x + \angle y + 50^\circ = 180^\circ$ (angle sum property in ΔPQR)

$\Rightarrow 2\angle x = 130^\circ$

$\Rightarrow \angle x = 65^\circ$

$\Rightarrow \angle y = 65^\circ$

(iii) Now, from Fig. 5.7(iii), in ΔABC $\angle A = x$ (vertically opposite angles)

Similarly $\angle B = \angle C = x$ (Why?)

$\Rightarrow \angle A + \angle B + \angle C = 180^\circ$ (angle sum property in ΔABC)

$\Rightarrow 3x = 180^\circ$

$\Rightarrow x = 60^\circ$

$\Rightarrow y = 180^\circ - 60^\circ = 120^\circ$

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